A Multi-Layer Perceptron Approach for SIF Retrieval in the O_2-A Absorption band from Hyperspectral Imagery of the HyPlant Airborne Sensor System

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Abstract

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Abstract

Accurate estimation of solar-induced fluorescence (SIF) from passively sensed hyperspectral remote sensing data has been identified as instrumental in assessing the photosynthetic activity of plants for various scientific and ecological applications at various spatial scales. Different techniques to derive SIF have been developed over the last decades. In view of ESA's upcoming Earth Explorer satellite mission FLEX aiming to provide high-quality global imagery for SIF retrieval an increased interest is placed in physical approaches. We present a novel method to retrieve SIF in the O_2-A absorption band of hyperspectral imagery acquired by the HyPlant sensor system. It aims at a tight integration of physical radiative transfer principles and self-supervised neural network training. To this end, a set of spatial and spectral constraints and a specific loss formulation are adopted. In a validation study we find good agreement between our approach and established retrieval methods as well as with in-situ top-of-canopy SIF measurements. In two application studies, we additionally find evidence that the estimated SIF (i) satisfies a first-order model of diurnal SIF variation and (ii) locally adapts the estimated optical depth in topographically variable terrain.

Keywords: Solar-Induced Fluorescence, Spectral Fitting Methods, Deep Learning, Neural Networks, Hyperspectral Sensors, Remote Sensing, SIF Retrieval

1. Introduction

The notion that sun-induced fluorescence (SIF) is an important biophysical parameter has become prevalent over the last decades [1]. In the last two decades, sensor and retrieval method development on the one hand and the push for the standardization of hyperspectral imagery in airborne and spaceborne sensing systems on the other hand have set the path to establish spatially contiguous SIF estimates [2, 3, 4, 5]. The close causal link of the SIF signal to the internal photosynthetic machinery of plants is a valuable tool to infer plant dynamics remotely over large areas. SIF has found applications linked to establishing regional and global gross primary productivity and light-use efficiency of vegetation [6, 7, 8], small and large scale quantification of photosynthetic dynamics [9, 10], the detection of various vegetation stress types [11, 12, 13], vegetative transpiration rates [14], and ecological monitoring more generally [15, 16, 17]. The retrieval of SIF from airborne hyperspectral imagery in particular allows to follow spatial and temporal patterns of photosynthetic activity typically at meter to submeter resolution. This provides important information in precision farming and phenotyping applications, and has been investigated in the context of food production [18].

A range of hyperspectral sensors on different platforms exist for operational passive retrieval of top-of-canopy SIF [19]. In the close range SIF retrieval, the use of field spectrometers has grown in recent years due to increased interest in passive SIF measurements as indicators of plant status in application studies. Furthermore, these ground measurements are instrumental in the development of UAV-based, airborne and spaceborne sensors for SIF retrieval. High-quality hyperspectral ground data sets for calibration and validation of non-ground-based sensors are increasingly needed [20] as the number of SIF products increases. In recent years, a multitude of globally distributed spaceborne SIF products have been developed from sensors that were originally designed for the characterisation of the atmosphere such as GOSAT [21], GOME [6, 22], SCIAMACHY [23], OCO-2 [24], TROPOMI [25, 26] and TanSAT [27]. In addition, the upcoming FLEX satellite mission will operationalize the first hyperspectral sensor dedicated full-spectrum SIF retrieval on a global scale by 2026 [28].

Investigations regarding the development, improvement and testing of SIF retrieval algorithms were conducted in the past mainly with data from airborne platforms equipped with sensors such as with the micro-hyperspectral imaging sensor [29], with the Chlorophyll Fluorescence Imaging Spectrometer [30] and with the HyPlant spectrometer [31]. The development of the new SIF retrieval approach presented in this work is based on hyperspectral HyPlant imagery. HyPlant data is particularly suitable to testing new retrieval methods as it has been designed as an

airborne demonstrator for the FLORIS sensor of the upcoming spaceborne FLEX mission. Moreover, hundreds of HyPlant flight lines were recorded in several field campaigns since 2014. This large collection of hyperspectral data sets allowed to test different SIF retrieval methods [32, 33, 34, 35, 36, 37, 38]. Many of these data sets are acquired in tandem with ground-based measurements allowing to derive high quality ground-based SIF estimates as 'ground truth' in addition to the hyperspectral flight lines (see Sec. 2.2 for details on the data sets and ground measurements).

Prominent algorithms for SIF retrieval on hyperspectral data are the 3FLD method [39], the Improved Fraunhofer Line Discrimination (iFLD) [40], the Spectral Fitting Method (SFM) [41, 42, 43] developed in view of the ambitious requirements of the FLEX mission, the Singular Vector Decomposition (SVD) method [44] as well as the recently developed partial least-squares (PLS) approach [45]. The present work will use SFM and iFLD versions as baseline methods that were specifically adapted to HyPlant data. Generally, such retrieval schemes aim at performing a decomposition of the at-sensor radiance into reflected solar irradiance and the top-of-canopy fluorescence transmitted to the optical sensor. This decomposition requires a joint estimation of the surface reflectance, the top-of-canopy fluorescence and the influence of atmospheric conditions during data acquisition, which cannot be inferred based on physical principles due to lacking knowledge of the atmospheric composition. It is an ill-posed inversion problem [46] that can be constrained by suitable assumptions on the different components. Fluorescence and reflectance are spectrally smooth, but vary strongly in space as they vary with surface properties. In contrast to this, atmospheric functions are spectrally highly variable but are spatially smooth as they vary only on much larger scales assuming constant atmospheric and observational conditions. Atmospheric transfer functions are therefore typically estimated for entire flight lines or large spatial image regions. To further disentangle fluorescence and reflectance iFLD and SFM formulate pixelwise functional constraints, i.e. constraints on full spectra or on a collection of spectral lines, that either allow to fit a specific at-sensor radiance model (SFM) or else to solve an equation system (iFLD). However, the assumption of constant atmospheric conditions causes performance decreases in regions with variable atmospheric transfer properties, e.g. because of varying optical path lengths due to complex topography. For this reason, the one of the main practical contributions of this study is that the new retrieval method considers locally varying atmospheric conditions and can thus can provide improved SIF estimates for topographically complex terrains.

We examine the performance of a spectral fitting method neural network (called SFMNN) trained with a novel self-supervised loss function specifically formulated for SIF retrieval from hyperspectral imagery first presented in [47]. Recently published

neural network based SIF retrieval schemes focus on reconstructing traditionally derived spaceborne SIF products from reflectance and PAR products and a subset of high-quality SIF estimates used as labels [48, 49, 50]. Our approach is different to these approaches as we do not use any labels for training, but leverage a set of constraints to invert a physical model of the at-sensor radiance. Our approach uses similar model assumptions as SFM for the different parts of the at-sensor radiance model, i.e. the fluorescence, reflectance, and atmospheric functions. However, a new approach to estimate the atmospheric transfer in HyPlant observations locally even in hilly terrain makes our approach a good candidate to extend the applicability of SFM to more diverse ecological domains of interest.

This work encompasses a qualitative comparison of the reconstruction residuals of SFMNN as compared to SFM and a quantitative comparison of its SIF predictions with FLOX top-of-canopy SIF measurements. Additionally, we show in two application studies that SFMNN's SIF predictions are in line with physiological and physical plausibility considerations.

In the first application study, we analyze the diurnal course of SFMNN predicted SIF. Understanding the diurnal dynamics of canopy SIF and its interplay with plant status and photosynthetic activity is an open and relevant research question [51, 52, 53, 54]. SIF as a function of daytime is has been modelled in the past by [55] and has been the subject of detailed studies more recently [56, 57]. As a plausibility check of SFMNN SIF predictions, we can make use of the universal feature of diurnal SIF dynamics to be strongly correlated to the local solar irradiance due to the large dependency of SIF on available photosynthetically active radiation (PAR). Given a predicted SIF time series we can formulate a first order plausibility test on whether the predicted SIF peaks around solar noon.

The SFMNN prediction of the atmospheric transfer is designed to be locally adaptive. The second application study focuses therefore on the sensitivity of the atmospheric transfer prediction to changing atmospheric and observational conditions. Due to a lack of knowledge about the atmospheric state at the time of data acquisition, we cannot compare the predicted atmospheric conditions to a measured or estimated atmospheric composition to gain insights into the physical accuracy of the predicted optical depth of the atmosphere. The only driver of the atmospheric transfer that can be quantitatively assessed by us is the optical path length of a given pixel on the ground. Therefore we test SFMNN's capacity to adapt to local changes in the optical path in HyPlant acquisitions with considerable topographic variation.

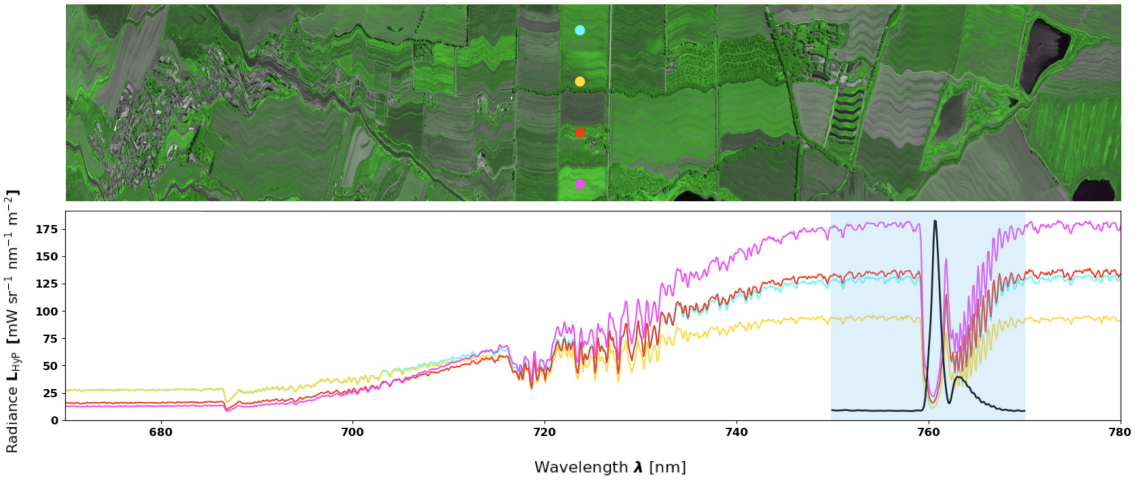


Figure 1: Sample spectra (coloured) of a WST-2019 HyPlant acquisition (upper row) over the input spectral window (lower row). The window $\mathcal{W}$ simulated by our approach is highlighted in blue. In black we show the SNR based weighting w_f described in Sec. 3.4.1.

2. Data

2.1. HyPlant Spectrometer

HyPlant is an assembly of two hyperspectral push-broom sensors with different spectral sampling and resolution configurations. The FLUO module covers the range 670 - 780 nm with a spectral sampling interval of 0.11 nm. Its resolution in terms of the full width half maximum (FWHM) is 0.27 nm [31]. See Fig. 1 for sample spectra over the full spectral range of HyPlant. The DUAL module in contrast, provides optical data in the spectral range 370 - 2500 nm with 3 - 10 nm spectral sampling and a FWHM varying between 1.7 - 5.6 nm. The FLUO module was specifically designed for SIF retrieval in the O_2-A and O_2-B absorption bands when operated on airborne platforms at flight heights varying from 350 m to 1800 m. The DUAL module's purpose is to simultaneously provide optical data from which ancillary information, such as the reflectance and various related products (e.g. NDVI), can be derived.

The current work makes use of radiometrically calibrated FLUO acquisitions in sensor geometry as input for the SIF predictor network [61]. The input data is not spectrally calibrated. Rather, the spectral calibration is directly built into the SIF retrieval scheme presented here as it is common for spectral fitting methods.

We train and predict SIF on data from 13 HyPlant campaigns in five locations from six years listed in Tab. 1. The chosen measurement campaigns cover a wide range of the nominal flight heights at which HyPlant is operated (350 - 1800 m). The

Campaign data set	FLOX	Δh [m]	HyP	DEM	Location
SEL-2018 (600 m)	✓	20	[36]		Selhausen, DE
SEL-2019 (600 m)	–	20	[37]		
SEL-2022 (600 m)	–	20	–		
WST-2019 (1500 m)	✓	20	[37]		Braccagni, IT
NRS-2019 (1500 m)	–	70	[37]	[58]	
TR32-2019 (1800 m)	–	150	[37]	[59]	Jülich, DE
INN-2022 (350 m)	–	60	–	[59]	
SOP-2023 (600 m)	–	140	–	[59]	
CKA-2020 (350 m)	✓	20	–		Klein
CKA-2020 (600 m)	✓	20	–		Altendorf, DE
CKA-2021 (350 m)	✓	20	–		
CKA-2022 (350 m)	✓	20	–		
GLO-2021 (1150 m)	✓	70	[38]	[60]	Mollerussa, ES

Table 1: Campaign data sets acquired in different locations in the years 2018 - 2023. With FLOX we denote the availability of simultaneous FLOX data, with Δh the maximum topographic variation over the campaign data set, with *HyP* the corresponding campaign report and with *DEM* corresponding the digital elevation map used for validation purposes. All campaign data sets and corresponding FLOX measurements are made available [61].

acquisitions of these measurement campaigns also differ with respect to the variability of the ground altitude. NRS-2019, TR32-2019, GLO-2019, INN-2022 and SOP-2023 comprise acquisitions with ground altitude changes of more than 50m, while WST-2019, SEL-2018 and the CKA campaigns were conducted over flat terrain. The variability of atmospheric transfer in the data sets, both within single acquisitions and between acquisitions, is large as a consequence.

As will be detailed below, the training process involves a pretraining and subsequent finetuning to compiled data sets consisting of HyPlant imagery acquired at the same nominal flight height and comparable topographic (and thus atmospheric) variation. For training, we crop the HyPlant imagery to windows of 17×17 pixels, called patches hereafter. The number of patches per compilation differs depending on data availability at individual flight heights. The selection of acquisitions for individual compilations ensured a balanced distribution of the sun-zenith angles (SZA) at which the acquisitions were taken. For the pretraining we compiled the data set

Data set		Included campaign data sets	$\lvert\mathcal{D}\rvert$
pretrain.	$\mathcal{D}_{\text{mix}}$	SEL-2018 (600 m), NRS-2019 (1500 m), WST-2019 (1500 m), CKA-2021 (350 m), GLO-2021 (1150 m)	260 (44)
flat	$\mathcal{D}_{350}$	CKA-2020, CKA-2021, CKA-2022	183 (44)
	$\mathcal{D}_{600}$	SEL-2018, SEL-2019, SEL-2020, SEL-2022, CKA-2019, CKA-2020, CKA-2022	283 (32)
	$\mathcal{D}_{1500}$	WST-2019	54 (16)
topo.	$\mathcal{D}^t_{600}$	INN-2022, SOP-2023	107 (12)
	$\mathcal{D}^t_{1150}$	GLO-2021	100 (8)
	$\mathcal{D}^t_{1500}$	NRS-2019	27 (12)
	$\mathcal{D}^t_{1800}$	TR32-2019	66 (8)
val.	$\mathcal{D}_{\text{day}}$	WST-2019	30 (9)

Table 2: Compiled data sets of HyPlant acquisitions from flight campaigns in the years 2018 - 2023. Flight heights are not mentioned if indicated in the data set name. In the first row, we indicate the purpose of the data set: *flat* or *topo.* denote the training of predictors for flat or topographically variable terrain, *val.* denotes specific validation purposes outlined in the Results section, *pretrain.* denotes the pretraining of a common predictor. $\lvert\mathcal{D}\rvert$ denotes the number of used 17x17 patches [$\times 10^3$] and in brackets the number of used acquisitions.

$\mathcal{D}_{\text{mix}}$ consisting of acquisitions with large differences in nominal flight height and topographic conditions. The variability of this data set ensures the generalizability to a wide variety of HyPlant campaigns during finetuning. For the finetuning process training data sets $\mathcal{D}_{350}$, $\mathcal{D}_{600}$ and $\mathcal{D}_{1500}$ were compiled by selecting a subset of the available acquisitions of the measurement campaigns with negligible topographic variation within single acquisitions (cf. *flat* in Tab. 2).

Parallelly, we created data sets with strong topographic variability (cf. *topo.* in Tab. 2). As for the *flat* denoted data sets, we grouped together campaign data sets acquired at the same flight height. Due to data limitations, the compiled *topo.* data

sets are smaller and constrained to single campaign site and acquisition years in all but one case.

The additional data set $\mathcal{D}_{\text{day}}$ was compiled for validation purposes. It is used both for validating SFMNN's reconstruction performance as well as the physiological plausibility of our approach in the first application study. It consists of nine repeated flights over the same location in an irregular frequency (10:11, 10:19, 10:27, 10:34, 13:15, 13:30, 13:38, 16:11, 16:18 CEST) allowing us to validate the performance of our approach under different solar zenith angles with otherwise comparable conditions. For our analysis of the SFMNN predicted diurnal course of SIF, we leverage the universal property of SIF to be strongly correlated to PAR under non-shaded conditions. This correlation is however only expected to be fulfilled over homogeneous fields. It may be invalid only under strongly changing irradiation in pixels with complex 3D structure. Under such conditions (i) varying sampling directions of the angular SIF emission distribution and (ii) shade impede the comparability of measurements at different times during the day. Predictions of the $\mathcal{D}_{\text{day}}$ data set are well suited for this analysis as the acquisitions have a relatively large pixel footprint (2.3 m) due to the large nominal flight height (1500 m). At this pixel resolution small scale inhomogeneities in crop fields are not preserved.

2.2. In-situ Fluorescence Measurements

A subset of the HyPlant measurement campaigns presented in Tab. 1 were accompanied by in-situ radiance and irradiance measurements recorded by the FLOX system (Fluorescence Box, JB-Hyperspectral Devices GmbH, Duesseldorf, Germany) [61]. The FLOX system provides precise radiance measurements on the ground within small footprints of a few square meters depending on the experimental setup. Fluorescence estimates derived from FLOX data sets have been used in the past for independent validation of HyPlant SIF estimates [62, 63].

The present study provides a systematic comparison of SFM, iFLD and SFMNN predictions from HyPlant imagery with FLOX derived SIF estimates. Data from top-of-canopy FLOX measurements from 2018 - 2022 were taken into account. We gained iFLD based FLOX SIF estimates with the processing software provided by the FLOX manufacturer. In order to validate with a consistent data set we disregarded FLOX measurements where the irradiance in the O_2-A band recorded at the beginning and at the end of a measurement cycle varied more than 1% because this is an indicator for changing illumination conditions. The majority of measurements were, however, taken under optimal measurement conditions so that only a negligible number of data points had to be disregarded.

Apart from the SEL-2018 FLOX data set, which was acquired with a mobile

FLOX system, the FLOX geolocation was acquired by a separate RTK-GPS measurement. The geolocation for the SEL-2018 FLOX data was derived from UAV data. Due to the high temporal variability of SIF, only FLOX data with an offset of up to ten minutes to a corresponding HyPlant overflight was considered. When multiple FLOX measurements were available within such a ten minute time window, we used the average for comparison with the HyPlant derived predictions, and the variance to compute an uncertainty estimate of the ground SIF measurement.

To account for spatial uncertainties associated with both the geolocation of the in-situ measurements as well as the field of view of single pixels in HyPlant FLUO data, we averaged the SIF prediction in a radius of 0.5m, 1m and 2m for acquisitions acquired 350, 600 and 1150 m above ground level, respectively, and used the variance within this region as an uncertainty estimate for the prediction. In addition, we used the variance of multiple FLOX derived SIF measurements collected within the chosen time window of ten minutes as an uncertainty estimate. Both the uncertainties determined for the FLOX measurements and the corresponding uncertainties calculated for the HyPlant derived SIF measurements were used to compute the uncertainty on the mean absolute error (MAE).

2.3. iFLD and SFM baseline SIF estimates

Both SFM and iFLD SIF estimates are generated operationally for all HyPlant campaigns. No dedicated rerun of the iFLD and SFM baseline methods was conducted for the comparisons presented in this work. Implementation details and application specifics of SFM and iFLD SIF retrievals applied to HyPlant FLUO data can be found in [31] and [64].

3. Development of a novel spectral fitting method

We propose a novel neural-network-based method to predict SIF in the O_2-A absorption band at 761 nm. It can be regarded as a spectral fitting method [65, 66] and we call it spectral fitting method neural network (SFMNN), accordingly. Similarly to other spectral fitting methods we first construct an explicit physical model of the measured at-sensor radiance which we then fit to HyPlant at-sensor radiance (L_{HyP}). The prediction of top-of-canopy SIF at 761 nm can be directly derived from the best estimates of the parameters modelling the top-of-canopy fluorescence emission curve. In contrast to typical spectral fitting methods such as SFM [43], where a least-squares minimization is performed, we implement a feature based optimization to fit the model to the observations. Specifically, we train a neural network (cf. Fig. 2) in a self-supervised fashion using the objective function minimized in SFM

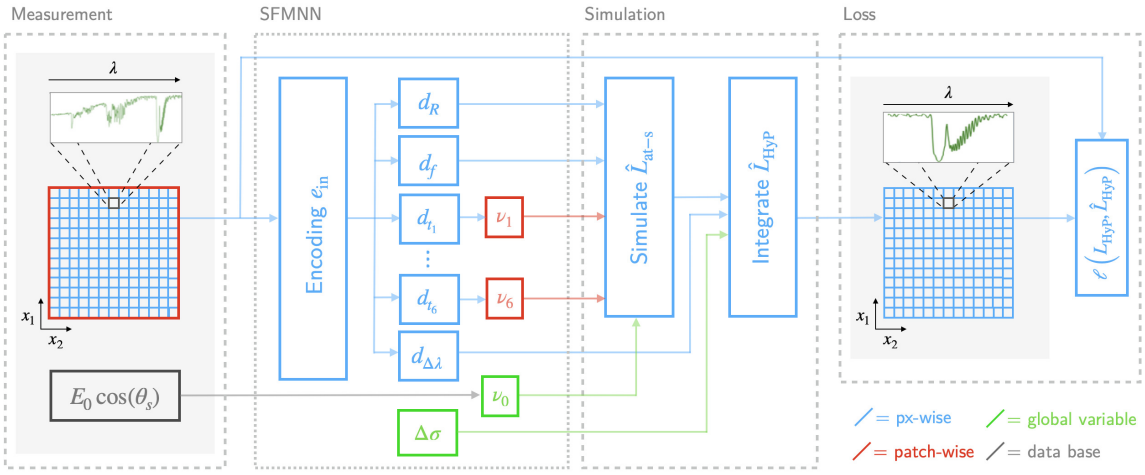


Figure 2: Outline of SFMNN's data input and architecture. Hyperspectral image patches are encoded (e_{in}) in a pixel-wise fashion to a common latent space. The latent vectors are subsequently decoded to physical variables (d_R, d_f, ...) parameterizing a four-stream model simulating the high-resolution at-sensor radiance $\hat{L}_{\text{at-s}}$. The patchwise decodings (d_{t_1}, d_{t_6}) are homogenised across the patch for that purpose. In a second simulation step, the estimated sensor characterization ($\Delta\lambda, \Delta\sigma$) is applied to $\hat{L}_{\text{at-s}}$ in order to yield a reconstructed observation $\hat{L}_{\text{HyP}}$. Different colors denote the range over which variables are kept constant (*px-wise* varying in every pixel, *patch-wise* over patches, *global-variable* over a single training data set). *data base* denotes the origin of the solar irradiance model [67] and solar zentih angle θ_s calculated for each acquisition.

as a part of the loss formulation. Self-supervised learning means that we do not need SIF ground truth for every training sample, as would be needed for supervised learning. Instead, we use the predicted SIF (and reflectance and atmosperic transfer functions) in a physical forward simulation model (the same as used in SFM) to reconstruct the signal. Comparing the reconstructed signal to the input signal allows to evaluate a signal reconstruction error as loss for usual backpropagation during training of the SFMNN. Training is thus completely independent of any SIF ground measurements; a fact allowing us to finetune the model to any new data even during prediction where needed or desired. However, differently to SFM, in SFMNN atmospheric transfer functions can be and are estimated locally over small patches which allows to account for changing observation and atmospheric conditions in a straightforward fashion. Moreover, SFMNN relies only on the full HyPlant at-sensor radiance spectrum and a general distribution of high-resolution atmospheric transfer functions.

3.1. Simulation of HyPlant at-sensor radiance

The loss formulation used for training relies on an explicit forward simulation model of the measured at-sensor signal L_{HyP} in the spectral range $\mathcal{W}_{\mathrm{tot}}$. In order to formulate a model for the physical at-sensor signal, we make use of the forward simulation formulation used in the SFM baseline [41], which is based on a four-stream radiative transfer model [68] extended by the addition of diffuse and direct fluorescence fluxes. It reads

$$\begin{aligned} L_{\mathrm{at-s}} = & \frac{E_s^0 \cos(\theta_s)}{\pi} \\ & \times \left(\rho_{so} + \tau_{oo} \left(\tau_{ss}\, r_{so} + \frac{\tau_{sd} + \tau_{ss}\, \overline{r_{sd}}\, \rho_{dd}}{1 - \overline{r_{dd}}\, \rho_{dd}}\, r_{do} \right) + \tau_{do}\, \frac{\tau_{sd}\, \overline{r_{dd}} + \tau_{ss}\, \overline{r_{sd}}}{1 - \overline{r_{dd}}\, \rho_{dd}} \right) \\ & + \tau_{oo}\, \frac{\left(f + \overline{f_d}\, \rho_{dd}\, r_{do} \right) + \overline{f_d}\, \tau_{do}}{1 - \overline{r_{dd}}\, \rho_{dd}}. \end{aligned} \tag{1}$$

We define all quantities involved in Eq. 1 in Tab. 3. After rewriting the functions related to atmospheric transfer as $t_0 \ldots t_6$ and products thereof as $t_7 \ldots t_{12}$ (cf. Tab. 3) and simplifying Eq. 1 by setting $r_{sd} = r_{do} = R$, $\overline{r_{sd}} = \overline{r_{dd}} = \overline{R}$, $\overline{f_d} = \overline{f}$ we get

$$\begin{aligned} L_{\mathrm{at-s}} &= L_{\mathrm{at-s}}^{R} + L_{\mathrm{at-s}}^{f} \\ &= t_0 \left(t_1 + t_7\, R + \frac{t_8\, R + \overline{R}\, R\, t_{12} + \overline{R}\, (t_9 + t_{10})}{1 - t_2\, \overline{R}} \right) \\ &\quad + \left(t_5\, f_s + \frac{\overline{f}\, (t_6 + t_{11}\, R)}{1 - t_2\, \overline{R}} \right) \end{aligned} \tag{2}$$

where the overline denotes a spatial mean. In practice this mean is computed by taking the mean over the output window. We estimate $t_1 \ldots t_6$, f and R at a high spectral resolution ($\delta\lambda = 0.0055$ nm) by modelling each of these functions in dependency of the input data. The implementation details of the modelling of each of these functions is detailed below in Sec. 3.2.1.

For the application of the neural network to the spectrally uncalibrated HyPlant radiance product L_{HyP}, the simulation model needs to take into account HyPlant's sensor characteristics in addition to the physical process which only yields $L_{\mathrm{at-s}}$. In order to simulate L_{HyP}, we need to estimate the sensor's center wavelength shift $\Delta\lambda$ and its spectral resolution $\Delta\sigma_\lambda$. Given an estimated center wavelength shift per pixel $\Delta\lambda$ and spectral resolution σ_λ, a simple sensor model can be created

$$\hat{L}_{\mathrm{HyP}}(\lambda, x, y) = \int_{\mathcal{W}} d\lambda\; g\left(\lambda + (\Delta\lambda)_{x,y} \mid \lambda_i,\, \sigma_\lambda \right)\; L_{\mathrm{at-s}}\left(\lambda + (\Delta\lambda)_{x,y} \right) \tag{3}$$

This model gives us the at-sensor radiance as measured by HyPlant $\hat{L}_{\mathrm{HyP}}(\lambda)$ in the spectral range $\mathcal{W}$ considered in the prediction. $\hat{L}_{\mathrm{HyP}}$ can be compared to radiometrically calibrated at-sensor radiance $L_{\mathrm{HyP}}(\lambda_i)$ in the nominal center wavelengths λ_i. In practice, we assume a Gaussian Spectral Response Function (SRF) g for all λ_i. This allows us to implement Eq. 3 approximately by Gaussian smoothing followed by a linear interpolation. Furthermore, we only estimate a shift $\Delta\sigma_\lambda$ with respect to the nominal spectral resolution of HyPlant $\langle\sigma_\lambda\rangle$, such that $\sigma_\lambda = \langle\sigma_\lambda\rangle + \Delta\sigma_\lambda$.

3.2. Architecture

SFMNN acts on patches of spatially contiguous input c_{xy}. The patch dimensions was fixed throughout all experiments to 17×17. This input consists of channelwise concatenation of the at-sensor radiance L_{HyP} bands. All parameters are predicted in a pixelwise fashion except parameters of the atmospheric functions t_i and $\Delta\sigma_\lambda$. SFMNN consists of a normalization layer, a dimensionality reduction layer k_{in}, an encoder e_{in} and several decoder modules for each of the variables. Both e_{in} and the decoder modules are simple multi-layer perceptrons with residual links.

The spatially dependent input c_{xy} passed to SFMNN is first normalized as presented in Fig. 3, where we show the first layers of SFMNN. The normalization layer is implemented as a batch normalization layer that is trained with the network during one epoch. In order to decrease the large input dimensionality of more than 1024 channels, we introduce a dimensionality reduction layer simply defined as

$$k_{\mathrm{in}} : \mathrm{ReLU} \circ \mathrm{BN} \circ \ell_{k_{i-1} \to k_i} \tag{4}$$

where $\ell_{k_{i-1} \to k_i}$ is a linear layer with input and output dimensions d_{in} and d, BN denotes a batch normalization layer and ReLU a rectified linear unit.

The encoder e_{in} and the decoder modules are defined in a very similar way. We define a module h as $k_{\mathrm{out}} \circ z_L \circ \cdots \circ z_0$ with

$$z_i(x) = \mathrm{D}_p^i \circ \mathrm{BN} \circ \mathrm{ReLU} \circ \left(\ell_{k_{i-1} \to k_i}(x) \; + (k_i \circ \cdots \circ k_i)\,(x)\right) \tag{5}$$

where D_p^i denotes a dropout layer with dropout probability p. k_i are defined as in Eq. 4. The dimensions of the linear layers of all submodules of e_{in} are listed in Tab. 4.

3.2.1. Mapping the decoded input spectra to physical parameters

Decoder modules are constructed identically to the encoder e_{in}, i.e. they are implemented as h modules but with submodules of different dimensions as compared to e_{in}. We list all the dimensions of the decoder module in Tab. 4. All decoders act on the same spatially dependent encoding $p_{xy} = e_{\mathrm{in}}(c_{xy})$ of the input c_{xy}. Each

Eq. 1	Eq. 2	Defnition
$E_0 \cos(\theta_s)$	t_0	solar irradiance modulated by the local solar zenith angle
f		directional fluorescence emission
$\overline{f_d}$	$\overline{f}$	diffuse fluorescence emission
ρ_{so}	t_1	bi-directional atmospheric reflectance
ρ_{dd}	t_2	spherical albedo at the bottom of the atmosphere
τ_{oo}	t_3	directional atmospheric transmittance along observation direction
τ_{ss}	t_4	directional atmospheric transmittance in solar direction
τ_{do}	t_5	directional atmospheric transmittance in observation direction for diffuse incidence
τ_{sd}	t_6	diffuse transmittance for solar incidence
r_{so}	R	bi-directional reflectance factor (BRF)
r_{do}	R	hemispheric-directional reflectance factor (HDRF)
$\overline{r_{dd}}$	$\overline{R}$	bi-hermispherical reflectance factor (BHRF) of the surrounding
$\overline{r_{sd}}$	$\overline{R}$	directional-hemispherical reflectance factor (DHRF) of the surrounding
$\tau_{ss}\,\tau_{oo}$	t_7	multiplication definitions in Eq. 2
$\tau_{sd}\,\tau_{oo}$	t_8	
$\tau_{ss}\,\tau_{do}$	t_9	
$\tau_{ss}\,\rho_{dd}$	t_{10}	
$\tau_{oo}\,\rho_{dd}$	t_{11}	
$\tau_{ss}\,\rho_{dd}\,\tau_{oo}$	t_{12}	

Table 3: Definitions for all quantities involved in the simulation model as defined in Eq. 1 and 2.

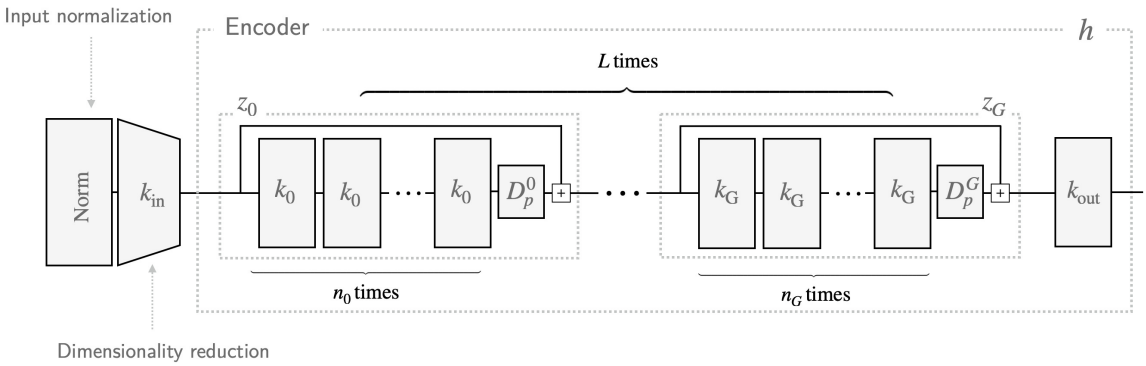


Figure 3: Conceptual sketch of the concatenation of input normalization layer, dimensionality reduction and encoder architecture h.

	k_i	n_i	D_p^i
Encoder e_{in}	(100, 100, 50)	(3, 3, 3)	(0.1, 0, 0)
Decoder d_v	(100, 50, 50, 50)	(3, 1, 1, 1)	(0, 0, 0)

Table 4: Architecture dimensionalities according to Fig. 3. Each element in a tuple denotes an architecture parameter for the i-th encoder or decoder layer ($0 \leq i \leq G$). k_i denotes the dimensionality of the i-th encoder (decoder) sublayers, n_i denotes the number of repetitions of sublayers, D_p^i denotes the dropout rate of the output of the i-th encoder (decoder) layer.

decoding $q_v = d_v(p_{xy})$ for variable v is then mapped to the physical parameters in Eq. 2. The mapping between the decoder outputs q_v and the physical values of variables that are predicted in a pixelwise fashion is implemented by a scaling layer

$$m(q \,|\, b_\ell, b_u) = b_\ell + \text{sigm}\,(q)\,(b_u - b_\ell) \tag{6}$$

where sigm denotes the sigmoid function. This formulation restricts the solution space to predefined parameter ranges (b_ℓ, b_u). These parameter ranges were chosen according to prior knowledge and are listed in Tab. 5.

	R	s_R	e	A_f	σ_f	$\Delta\lambda$	E	t
Lower bound b_ℓ	0	0	0	0	20	-0.09	0.98	-300
Upper bound b_u	0.6	12 $\times 10^{-4}$	1	8	21	0.09	1.02	300

Table 5: Upper and lower bounds parameterizing the scaling layers for each variable needed in the signal reconstruction.

We allow different $\Delta\lambda$ across acquisitions and different campaigns. We include this variation in the forward simulation by constructing a decoder $d_{\Delta\lambda}$ with a scaling layer

$$d_{\Delta\lambda}(p) = m\left(d(p) \,|\, (b_{\Delta\lambda})_\ell\,,\, (b_{\Delta\lambda})_u\right) \tag{7}$$

Similarly, we construct the decoder $d_f(p) \rightarrow (A_f, \sigma_f)$ that yields the parameterization for the assumed functional shape for fluorescence emission

$$f(\lambda) = A_f\, \mathcal{N}\left(\lambda \,|\, \mu_\lambda, \sigma_f\right) \tag{8}$$

around the emission peak $\mu_f = 740$ nm. The prediction bounds parameterizing the scaling layer are again given in Tab. 5. Since the present contribution is restricted to SIF estimation in the O_2-A absorption band, we restrict f to the emission peak at 740 nm.

Equivalently we restrict the definition of the reflectance R to the spectral range around the O_2-A absorption band (740 - 780 nm). We assume R can be approximately expressed by a square polynomial in this spectral range. Following [69, 70] a network $d_R(p) \rightarrow (\rho, s_\rho, e)$ is constructed in a way that reflectance estimates can be written

$$R(\lambda \,|\, p) = \rho \,+\, s_\rho\,(\lambda - \lambda_0) \,+\, \frac{s_\rho\,(e-1)(\lambda - \lambda_0)}{2\,(\lambda_1 - \lambda_0)} \tag{9}$$

where we define $\lambda_0 = 740$ nm and $\lambda_1 = 780$ nm. The bounds of the scaling layer for ρ, s_ρ and e are given in Tab. 5. It has proven to allow a physiologically consistent parameterization of observed reflectance spectra.

Lastly, we highlight that we estimate a single FWHM shift $\Delta\sigma_\lambda$ for the whole training data set, contrarily to the CW shifts $\Delta\lambda$ and fluorescence and reflectance parameters which are estimated for each pixel. We chose this simplification of the retrieval problem as preliminary analysis has shown that any larger degree of freedom in the prediction of $\Delta\sigma_\lambda$ negatively impacted the fluorescence retrieval.

3.2.2. Atmosphere Estimation

We predict the atmospheric functions $t_1, \ldots, t_6$ and calculate $t_7, \ldots, t_{12}$ as the products as which they were defined in Eq. 2. Differently to SFM, we do not rely directly on MODTRAN simulations to estimate the atmospheric transfer. Instead, we perform a Principal Component Analysis (PCA) decomposition on a chosen set of MODTRAN simulated atmospheric functions that were derived by SFM for a range of different HyPlant acquisitions (cf. Fig. 4). Since we assume that we can find fitting

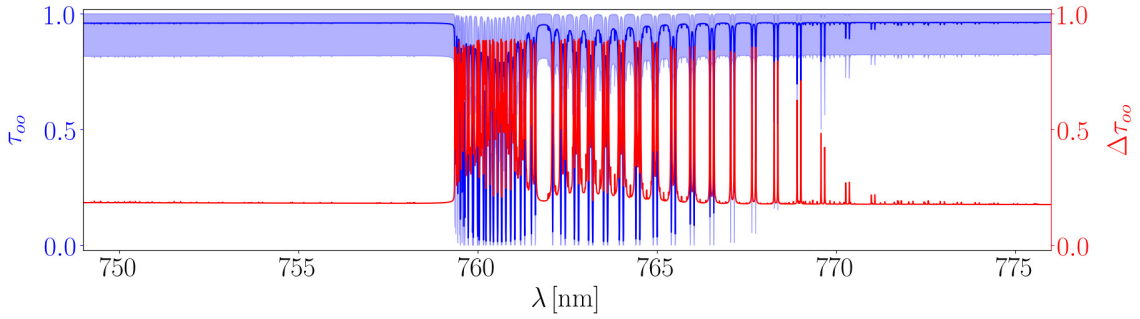


Figure 4: Distribution of the SFM estimated transfer function τ_{oo} used for the PCA decomposition. We show the lower and upper bound as well as the mean τ_{oo} in blue. The total extent of $\Delta\tau_{oo} = \max(\tau_{oo}) - \min(\tau_{oo})$ in the selected data set is shown in red.

atmospheric functions for arbitrary HyPlant acquisitions in the span of these PCA components, this PCA decomposition provides us with a suitable parameterization of the atmospheric transfer. This assumption is mainly conditioned on the fact that HyPlant campaigns are only operated under optimal weather conditions in a few regions. The variation in the atmospheric state is therefore assumed to be small.

In order to derive a suitable PCA decomposition we used high resolution (0.0055 nm) atmospheric transfer functions calculated during SFM optimization for 38 acquisitions from a compilation of acquisitions from 2018 and 2019 [36, 37]. Fig. 4 shows as an example the distribution of the SFM estimated transfer function τ_{oo} in these selected acquisitions. These acquisitions were chosen to cover a multitude of flight heights, flight directions, and day times to ensure that the span of the PCA components covers all acquisitions considered in this work. We highlight that we included atmospheric transfer functions of SEL-2018 and WST-2019 acquistions, which are also used for training and validation (cf. Tab. 1) in this data set. In the case of acquisitions from these two campaigns we thus guarantee that the SFM solution for the atmospheric transfer functions is in the solution space of SFMNN's atmospheric predictor.

We derived the first five principal components c_{ij} from the distributions of $t_1, \ldots, t_6$ before training the network. These components c_{ij} (with $1 \leq i \leq 6$ and $0 \leq j \leq 5$) and according means μ_i and standard deviations σ_i were then used for all experiments. In order to estimate PCA weights from the encoded HyPlant radiance input p, we construct networks $d_{t_i}(p) \rightarrow \{q_{ij}\}_{j \leq N_c}$ such that

$$t_i(p) = \min\left(1, \mu_i + \sigma_i \sum_{j \leq N_c} c_{ij} w_{ij}\right), \text{ with } w_{ij} = m\left(q_{ij} \mid (b_t)_\ell, (b_t)_u\right) \tag{10}$$

where p is the single pixel encoding as defined above and N_c is the number of components used for the PCA reconstruction.

For the solar top-of-atmosphere irradiance t_0 we simply predict a single weight q_0 for the whole training dataset and use the constant solar irradiance E_s^0 and the true solar zenith angle θ_s corresponding to the acquisition time and location as initial guess from which the network is allowed to deviate slightly, i.e.

$$t_0 = E_s^0 \cos(\theta_s) \left((b_E)_\ell + \mathrm{sigm}(q_0) \left((b_E)_u - (b_E)_\ell \right) \right). \tag{11}$$

Here, q_0 is a free parameter and does not depend on the spectral input p since we introduce possible deviation from the solar constant E_s^0 only to account for possible miscalibration of E_s^0. Since t_1 is supposed to be constant, the bounds $(b_E)_\ell$ and $(b_E)_u$ are chosen tightly.

3.3. Architectural constraint

The architecture of $d_{t_i}(p)$ differs from the decoder structure introduced before, because we require a single prediction t_i per patch and not per pixel. This is necessary since we architecturally constrain the ill-posed retrieval by requiring the atmospheric prediction to hold over an extended spatial domain. We implement this by first averaging a per-pixel atmosphere encoding $e_{t_i}(p)$ over the patch. $e_{t_i}(p)$ is architecturally identical to d_f and d_R. Thus, we formulate $d_{t_i}(p) = q_i$ as

$$q_i = \nu_i \left(p_{xy} \right) = \mathrm{BN} \circ \ell_{d_a \to N_c} \circ \sum_{x,y \leq s} s^{-2} \, e_{t_i}(p_{xy}) \tag{12}$$

where $\ell_{d_a \to K}$ is a linear layer with input and output dimensions d_a and N_c and where BN denotes a batch normalization layer.

3.4. Loss & Regularization

We formulate the following unsupervised loss per patch for the SIF regression problem for the spectral domain $\mathcal{W}$

$$\ell(L_{\mathrm{HyP}}, \hat{L}_{\mathrm{HyP}}) = (\ell_{\mathcal{W}} + \gamma_f \ell_f + \gamma_N \ell_{\mathrm{NDVI}} + \gamma_a \ell_{\mathrm{atm}})(L_{\mathrm{HyP}}, \hat{L}_{\mathrm{HyP}}). \tag{13}$$

This loss implements a least-squares minimization on the measured at-sensor radiance. It is composed of the main loss contribution over the spectral range $\mathcal{W}$

$$\ell_{\mathcal{W}} = |\mathcal{W}|^{-1} \, \mu \left(\sum_{\lambda \in \mathcal{W}} \left(L_{\mathrm{HyP}}(\lambda) - \hat{L}_{\mathrm{HyP}}(\lambda) \right)^2 \right), \tag{14}$$

where μ denotes the mean over all pixels. The three additional signal ℓ_f, physiological ℓ_{NDVI} and physical ℓ_{atm} regularizers account for data-specific constraints.

3.4.1. Signal regularization

The regularizer

$$\ell_f = |\mathcal{W}|^{-1}\, \mu \left(\sum_{\lambda \in \mathcal{W}} w_f(\lambda) \left(L_{\text{HyP}}(\lambda) - \hat{L}_{\text{HyP}}(\lambda) \right)^2 \right)_{\delta \text{R} = \delta \text{t}_{\text{atm}} = 0} \tag{15}$$

weights the reconstruction with a sensor and signal specific weight function w_f. w_f boosts the loss in spectral regions with high fluorescence SNR. The gradient computation of ℓ_f is set to only influence the fluorescence module d_f as the definition of w_λ assumes known reflectance and atmospheric transfer.

We define w_f as the Moore-Penrose solution to a simplified and linearized retrieval problem. The derivation of the linearized problem and these weights (cf. Sec. Appendix A) result in the definition

$$w_f(\lambda) = \left\langle \frac{1}{\sigma_\lambda^2} \frac{\sum_{\lambda'} f_{\lambda'}^2}{\sum_{\lambda'} f_{\lambda'}^2 / \sigma_{\lambda'}^2} \right\rangle_{p(L_{\text{at-s}}, f)} \tag{16}$$

where f denotes the predicted at-sensor fluorescence, σ_λ the estimated sensor variance of the total at-sensor radiance and $\langle \dots \rangle_{p(L_{\text{at-s}}, f)}$ the mean over a representative distribution of at-sensor radiance and fluorescence emission. We show the mean weights w_f used in all experiments that we estimate using a fixed distribution $p(L_{\text{at-s}}, f)$ in Fig. 1. Both the variability of w_f as well as the influence of w_f on the learned retrieval have not been investigated in this work.

3.4.2. Physiological constraint

The regularizer

$$\ell_{\text{NDVI}} = \mu \left(\hat{f}\, \delta \left(\text{NDVI} \leq \tau_{\text{NDVI}} \right) \right) \tag{17}$$

was introduced to reinforce the physiological expectation of vanishing fluorescence in sparsely vegetated areas. We fixed $\tau_{\text{NDVI}} = 0.15$ in all experiments in accordance with a similar thresholding principle in SFM [43]. The NDVI is derived online during training and inference from the FLUO input spectra.

3.4.3. Physical Regularization

For practical purposes, we define the total predicted transfer function

$$t_{\text{tot}} = \frac{L^R_{\text{at-s}}\,\overline{R}^{-1}}{E^0_s\,\cos(\theta_s)} \tag{18}$$

describing both downwelling, upwelling and diffusive contributions to the at-sensor signal. In this computation, we only use the reflectance signal contribution $L^R_{\text{at-s}}$ from Eq. 2.

We point out that the resulting effective transfer function t_{tot} may not respect $\max(t_{\text{tot}}) < 1$ because of its diffusive contributions even though individual components $t_1 \dots t_6$ are constrained to < 1. Since we can assume diffusive transfer to be small at normal acquisition conditions during HyPlant campaigns we make sure that the atmospheric transfer functions result in a physically plausible total transfer function order by regularizing the loss with a soft constraint enforcing $t_{\text{tot}} \lesssim 1$. Thus, we add

$$\ell_{\text{atm}} = \text{ReLU}(t_{\text{tot}} - 1) \tag{19}$$

as an additional regularizer to the loss. It constrains the network to decrease diffusive parts of the predicted atmospheric transfer. While this regularizer achieves the soft constraint $t_{\text{tot}} \lesssim 1$ with suitable weight γ_a in the loss (cf. Eq. 13) it has the downside that γ_a effectively controls the contribution of simulated diffusive at-sensor radiance.

3.5. Training Set Up

The training of the SFMNN network was conducted in two phases. First we performed a pretraining on a data compilation of acquisitions from different nominal heigths $\mathcal{D}_{\text{mix}}$ (cf. Tab. 1). This pretraining provided us with a coarsely trained network p_{mix} which we then finetuned to gain predictors for specific observation and terrain conditions. We trained networks p_{350}, p_{600} and p_{1500} for individual nominal flight heights (350, 600 and 1500 m) in flat terrain. In Tab. 1 we detail the composition of the corresponding compilations $\mathcal{D}_{350} \dots \mathcal{D}_{1500}$ that were used for fine tuning. In addition to the flat terrain predictors, we also compiled data sets with variable topography ($\mathcal{D}^t_{600}$, $\mathcal{D}^t_{1150}$, $\mathcal{D}^t_{1500}$ and $\mathcal{D}^t_{1800}$). Due to a significantly reduced availability of HyPlant data in topographically complex terrain these compilations mainly consist of acquisitions from a single campaign.

The HyPlant acquisitions in a training data set were cut spatially in partially overlapping patches. A fixed patch size of 17 × 17 HyPlant pixels was chosen. While we used the whole spectral range covered by the FLUO module as spectral input window $\mathcal{W}_{\text{in}}$, we restricted the prediction range to $\mathcal{W} = [750, 770]$ nm. In all

training runs, the input provided to the network consisted of concatenated layers of the spectral information, the across-track pixel location as well as the mean solar incidence angle per acquisition. It is noteworthy that no topographic elevation such as a digital elevation model (DEM) was provided to SFMNN during training or testing time.

The training was conducted with an Adam optimizer guided gradient descent. The nominal learning rate was set initially to $\ell_r = 10^{-3}$ and subsequently reduced to a minimum of $\ell_r = 10^{-4}$ with a learning rate scheduler. We fixed the loss weights in Eq. 13 to $\gamma_f = 5$, $\gamma_N = 10$ and $\gamma_a = 1$. All networks (p_{mix}, $p_{350}\dots$) were trained until the validation loss on a random subset of the training data stopped decreasing.

4. Results

In order to validate the SFMNN reconstruction and retrieval performance, we present a qualitative comparison of the reconstructed at-sensor radiance of SFM and SFMNN as well as a quantitative comparison of SFMNN top-of-canopy fluorescence estimates with top-of-canopy FLOX measurements. Additionally, we include two application studies to assess the physiological and physical plausibility of SFMNN's estimated parameters.

4.1. At-Sensor radiance reconstruction performance

Good reconstruction performance of the at-sensor radiance is critical for successfully disentangling R, f and the atmospheric transfer functions t_i. In order for SFMNN to yield consistent SIF estimates, comparable reconstruction performance across a wide range of observational conditions is needed. We show that SFMNN reaches that goal by analyzing error statistics of the predictions in the O_2-A absorption band in Fig. 5, where we plot the 5% - 95% percentiles of the residual distributions of reconstructed radiance spectra in the spectral range $\mathcal{W}_{\text{O}_2-\text{A}} = [759.55, 761.60]$ nm. The SFMNN simulations exhibit root mean square error ($\text{RMSE}_{\mathcal{W}_{\text{O}_2-\text{A}}}$) distributions constrained consistently in the range 0.6 - 2 mW nm^{-1} sr^{-1} m^{-2}. We equally show the normalized residual

$$\text{nRMSE} = \text{RMSE}_{\mathcal{W}_{\text{O}_2-\text{A}}} / L_{\text{HyP}}(761\,\text{nm}) \tag{20}$$

in Fig. 5. Contrarily to the RMSE, we find increasing nRMSE with increasing SZA. The RMSE decreases more slowly than the mean at-sensor radiance with increasingly diffusive conditions for image data acquired under high SZA. As a consequence, the nRMSE increases under low-light conditions.

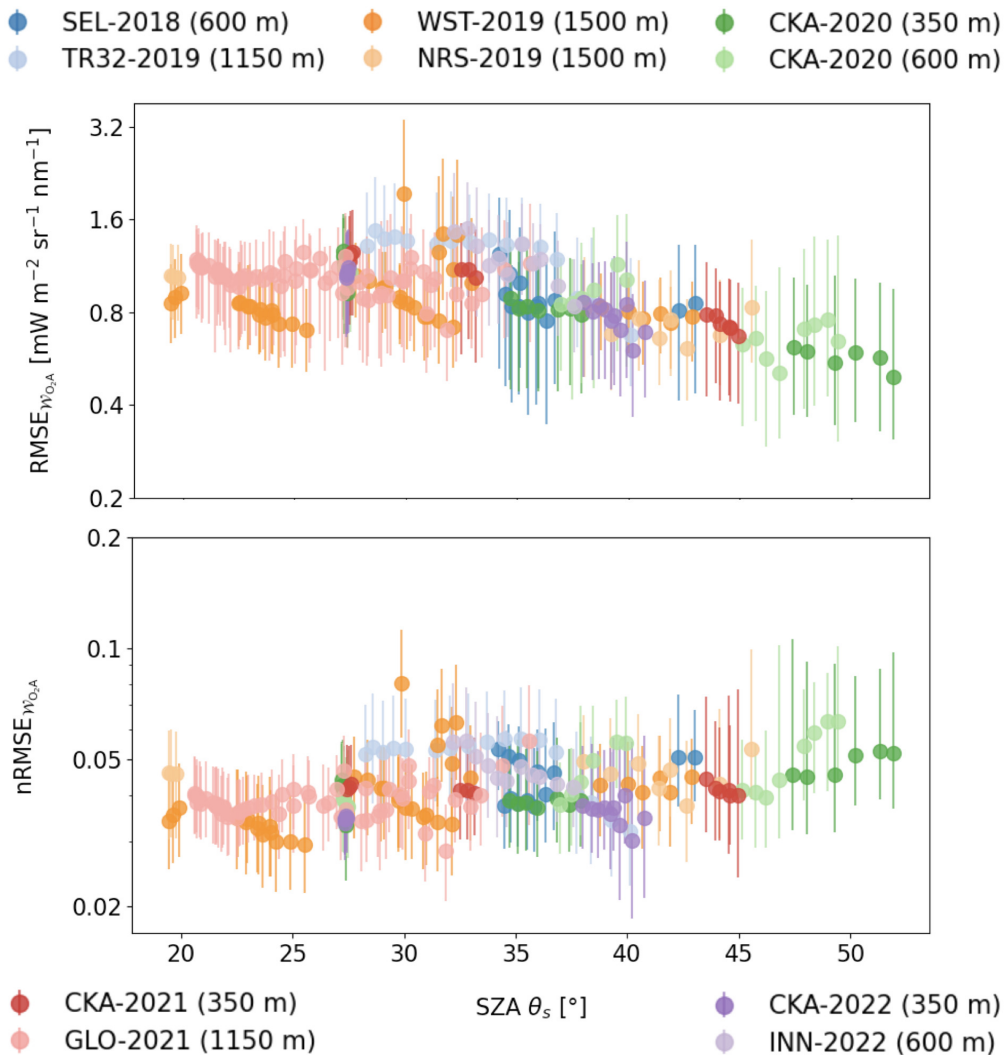


Figure 5: RMSE (upper row) and normalized RMSE (lower row) between observed radiance and radiance predicted by SFMNN in the spectral range $\mathcal{W}_{\mathrm{O_2-A}} = [759.55, 761.60]$ nm as a function of the sun zenith angle (SZA). Each data point corresponds to an acquisition to which we applied our SIF retrieval. Error bars cover the quantile range of 5 - 95 % within a single acquisition.

In order to assess whether SFMNN's reconstruction error is sufficient for a spectral fitting based SIF retrieval, we compare the reconstruction residuals of SFMNN and the SFM baseline. The present study only focuses on a comparison in the $\mathcal{D}_{\mathrm{day}}$ dataset. To highlight the variability in reconstruction performance across different light regimes we differentiate between *noon* acquisitions taken around the solar noon

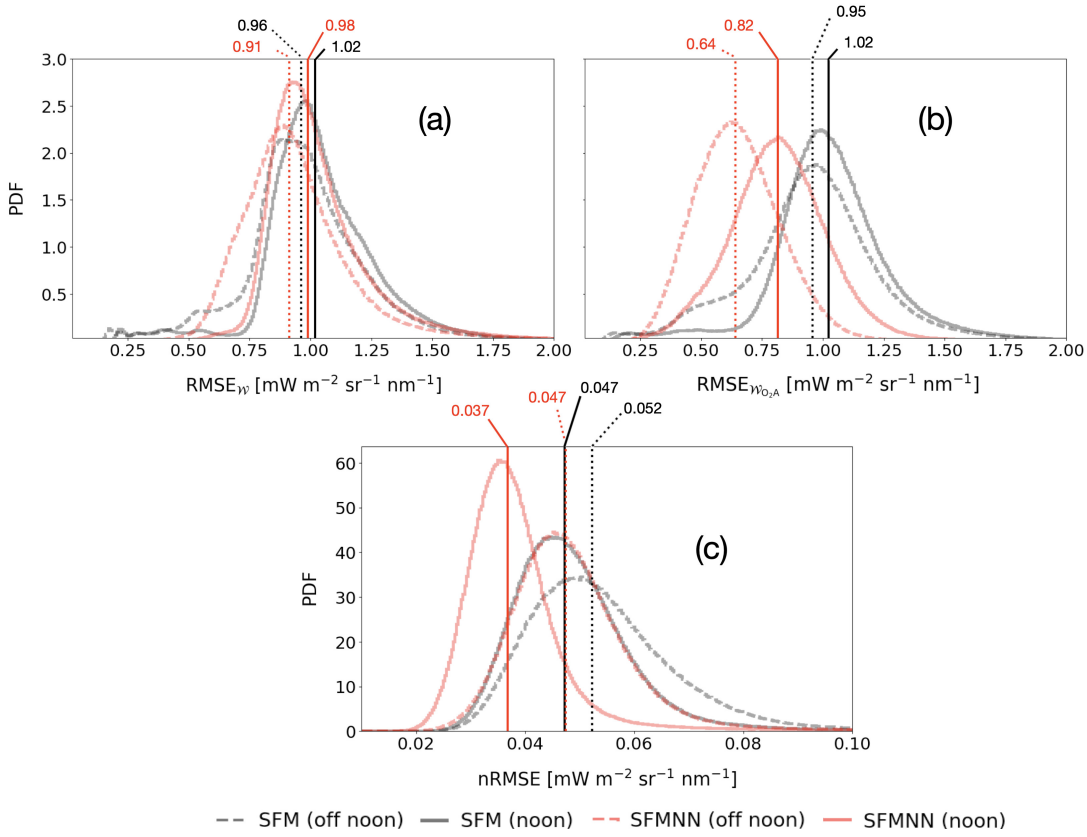


Figure 6: Residual distribution of SFM and SFMNN reconstructions of WST-2019 (1500 m) acquisitions sampling a single day (2019/06/19). *noon* denotes data from 12:00 - 14:00 (3 acquisitions), *off-noon* denotes data acquired earlier than 12:00 or later than 14:00 (6 acquisitions). Shown are (a) RMSE$_{\mathcal{W}}$ over the complete spectral window $\mathcal{W}$, (b) RMSE$_{\mathcal{W}_{O_2-A}}$ in the O_2-A band [759.55, 761.60] nm and (c) nRMSE as defined in Eq. 20.

(13:15, 13:30, 13:38 CEST) and *off-noon* acquisitions taken in the morning and afternoon (10:11, 10:19, 10:27, 10:34, 16:11, 16:18 CEST).

In Fig. 6 (a) and (b) we plot the residual distributions RMSE$_{\mathcal{W}}$ over the whole predicted spectrum $\mathcal{W} = [750, 770]$ nm and RMSE$_{\mathcal{W}_{O_2-A}}$ in the O_2-A band. SFMNN's reconstruction performance over the full spectrum (cf. Fig. 6 (a)) as well as within the O_2-A band (cf. Fig. 6 (b)) is improved over SFM. In both cases, we find lower reconstruction errors in *off-noon* than in *noon* acquisitions. The improved performance of SFMNN over SFM in the $\mathcal{D}_{\text{day}}$ can furthermore be observed in terms of nRMSE in 6 (c). As was established previously in the discussion of Fig. 5, the nRMSE increases in *off-noon* acquisitions with larger SZA both in SFMNN and SFM reconstructions.

We conclude that SFMNN is able to learn a reconstruction of the at-sensor radiance with a reconstruction error in the O_2-A band consistently bounded in the range 0.6 - 2 mW nm^{-1} sr^{-1} m^{-2} over the whole range of observational conditions covered by the data sets. SFMNN's prediction accuracy in the O_2-A band is improved over the operational SFM predictions in the $\mathcal{D}_{\text{day}}$ dataset both in terms of RMSE and nRMSE.

4.2. Validation of SFMNN predicted top-of-canopy fluorescence with FLOX measurements

In order to evaluate whether SFMNN's decomposition of the at-sensor radiance yields SIF estimates comparable to SFM and iFLD baselines, we directly validate SFMNN SIF with in-situ SIF measurements that were acquired quasi-simultaneously to the airborne HyPlant acquisitions. In particular, we compare FLOX derived iFLD SIF estimates to HyPlant derived SIF predictions for the five measurement campaigns SEL-2018, WST-2019 (1500 m), CKA-2020 (350 m), CKA-2020 (600 m), CKA-2021 (350 m), GLO-2021 (1150 m) and CKA-2022 (600 m) (cf. Tab. 2).

In Tab. 6 we show results taking into account all in-situ FLOX measurements available after filtering as described in Sec. 2.2. In addition to the MAE, the results report the Pearson correlation r.

The validation results show that SFMNN outperforms the two baseline methods in terms of r for all but two datasets (cf. Tab. 6). In these two data sets providing a lower validation accuracy, CKA-2020 (350 m) and CKA-2021 (350 m), r is still close to the performance of SFM and iFLD. A similar conclusion can be drawn when comparing the performance in terms of mean absolute error (MAE). SFMNN is among the best performing predictors in terms of MAE for all but one dataset (CKA-2021 (350m)).

The results for the CKA-2020 (350 m) and (600 m) runs refer to data acquired by three different FLOX systems placed in two different crop types (winter wheat, oat). In order to avoid biased metrics due to sensor specific and crop type phenological differences that potentially influenced the FLOX measurements, we additionally list the prediction performance over the measurement series of the individual measurement stations at the ground (cf. Tab. 7). The results in Tab. 7 show that there is some variation in the performance metrics across the different FLOX measurement set-ups. In the case of SFMNN, the variation in MAE is not larger than the estimated uncertainty associated with these MAE values. This observation also approximately holds for the SFM results. This performance consistency across different measurement locations is an indication that the SFMNN and SFM performances for those SIF point measurements can be generalized within the given uncertainty ranges.

Data Set		r	MAE	N
SEL-2018 (600m)	SFM	0.89	0.80 ± 0.10	11
	SFMNN	**0.96**	**0.68 ± 0.08**	11
	iFLD	0.80	**0.67 ± 0.08**	10
WST-2019 (1500m)	SFM	-0.36*	0.46 ± 0.05	22
	SFMNN	**0.62**	**0.19 ± 0.03**	22
	iFLD	-0.59	4.81 ± 0.09	22
CKA-2020 (350m)	SFM	**0.90**	0.36 ± 0.04	37
	SFMNN	0.87	**0.31 ± 0.04**	37
	iFLD	0.55	**0.28 ± 0.05**	36
CKA-2020 (600m)	SFM	**0.83**	0.42 ± 0.05	23
	SFMNN	**0.83**	**0.24 ± 0.05**	23
	iFLD	0.52	**0.39 ± 0.08**	23
CKA-2021 (350m)	SFM	0.64*	0.44 ± 0.07	7
	SFMNN	0.82	0.49 ± 0.08	7
	iFLD	**0.87**	**0.12 ± 0.15**	7
CKA-2022 (350m)	SFM	0.57*	**0.39 ± 0.13**	6
	SFMNN	**0.69***	**0.33 ± 0.16**	6
	iFLD	-0.88*	1.13 ± 0.21	4
GLO-2021 (1150m)	SFM	0.89	**0.24 ± 0.09**	6
	SFMNN	**0.92**	**0.28 ± 0.14**	6
	iFLD	0.81*	0.74 ± 0.16	5

Table 6: FLOX derived SIF measurements compared to SFMNN, SFM and iFLD SIF predictions ($\leq$ 10 min time difference). Correlation r marked with * have $p > 0.05$. Mean absolute errors (MAE) are given in mW nm^{-1} sr^{-1} m^{-2}. N denotes the number of validation points. We highlight the best results in bold font. Multiple MAE results are highlighted, if the uncertainty is too large to establish a single *best* case. All uncertainties are computed from the variation of HyPlant derived SIF estimates within a fixed radius (350 m: 0.5m, 600 m: 1m, 1150 m: 2m) around the measurement location and the variation of FLOX SIF estimates within the predefined time window of 10 min.

As a conclusion we highlight that the performance metrics of SFM and SFMMN could be considered similar, while the iFLD method provided SIF estimates with a distinctly lower accuracy. SFMNN slightly outperformed the state-of-the-art SFM SIF in terms of accuracy. Furthermore, a comparison of the performance of individual and aggregated FLOX performance metrics indicated that both SFM and SFMNN's performance can be generalized across the different measurement set-ups across and within the single campaigns.

Data		FLOX	r	MAE	N
CKA-2020 (350 m)	SFM	Wheat (1)	**0.62**	0.48 ± 0.05	11
		Wheat (2)	0.86	**0.29 ± 0.07**	14
		Oat	**0.85**	0.32 ± 0.06	12
		All	**0.90**	0.36 ± 0.04	37
	SFMNN	Wheat (1)	0.60	**0.32 ± 0.06**	11
		Wheat (2)	**0.90**	0.38 ± 0.08	14
		Oat	0.41*	**0.21 ± 0.06**	12
		All	0.87	**0.31 ± 0.04**	37
	iFLD	Wheat (1)	-0.19*	**0.33 ± 0.07**	11
		Wheat (2)	0.11*	**0.34 ± 0.09**	13
		Oat	0.10*	**0.17 ± 0.09**	12
		All	0.55	**0.28 ± 0.05**	36
CKA-2020 (600 m)	SFM	Wheat (1)	**0.77***	0.85 ± 0.09	4
		Wheat (2)	0.86	0.32 ± 0.10	10
		Oat	0.73	**0.33 ± 0.06**	9
		All	0.83	0.42 ± 0.05	23
	SFMNN	Wheat (1)	0.71*	**0.17 ± 0.08**	4
		Wheat (2)	**0.91**	**0.22 ± 0.10**	10
		Oat	**0.74**	**0.29 ± 0.06**	9
		All	**0.83**	**0.24 ± 0.05**	23
	iFLD	Wheat (1)	-0.52*	0.51 ± 0.10	4
		Wheat (2)	0.25*	0.45 ± 0.16	9
		Oat	0.07*	**0.26 ± 0.10**	9
		All	0.52	0.39 ± 0.08	23

Table 7: Performance metrics for different 3 different FLOX devices. We denote the different devices by the crop type the devices were pointed at. Metric definitions as in Tab 6. Correlation scores r marked with * have $p > 0.05$. Mean absolute errors (MAE) are given in mW nm^{-1} sr^{-1} m^{-2}. We highlight the best results in bold font. N denotes the number of validation points.

4.3. Application study 1: Retrieval of SIF in the diurnal course

In this application study we analyze the diurnal course of SFMNN predicted SIF and assess whether it peaks around the solar noon. We formulated this plausibility test on the diurnal data set $\mathcal{D}_{\text{day}}$, requiring that the second order derivative of the resulting SIF predictions be negative (cf. Fig. 7).

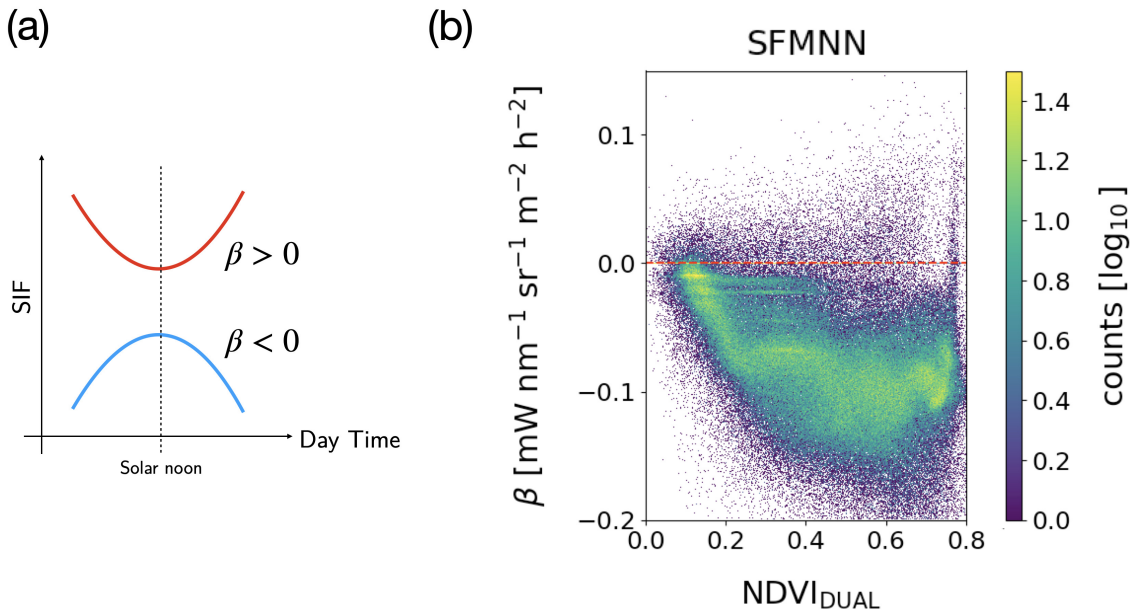


Figure 7: (a) sketch of relationship betwenn β and the diurnal variation in SIF, (b) histogram of SFMNN β-NDVI relationship shown in Fig. 8.

The SIF predictions for the nine HyPlant acquisitions in $\mathcal{D}_{\text{day}}$ were georegistered to form time series in each pixel. We detrended each time series independently and fitted a second order polynomial to each time series in each georegistered pixel. The second order derivative β then was used as an indication whether the predicted SIF peaks at solar noon ($\beta < 0$) or in morning and evening ($\beta \geq 0$) acquisitions (cf. Fig. 7 (b)). As would be expected for physiologically correct SIF estimates, we found the resulting β to be implicitly bounded by $\beta = 0$ over the whole NDVI domain. The spatial β distribution for $\mathcal{D}_{\text{day}}$ in Fig. 8 highlights that pixels with $\beta > 0$ are mainly distributed parallel to tree lines where the strong shadowing in the diurnal course invalidates the model assumptions. SFMNN predictions thus yield a plausible diurnal course of SIF.

4.4. Application study 2: Retrieval of SIF in hilly terrain

We have trained SFMNN p^t_{600}, p^t_{1150}, p^t_{1500} and p^t_{1800} on the data sets $\mathcal{D}^t_{600}$, $\mathcal{D}^t_{1150}$, $\mathcal{D}^t_{1500}$ and $\mathcal{D}^t_{1800}$ exhibiting large topographic variation. In this application study we show that the constraints imposed on SFMNN's weight optimization yield topography-aware predictors. For this we compare the optical path length d to the

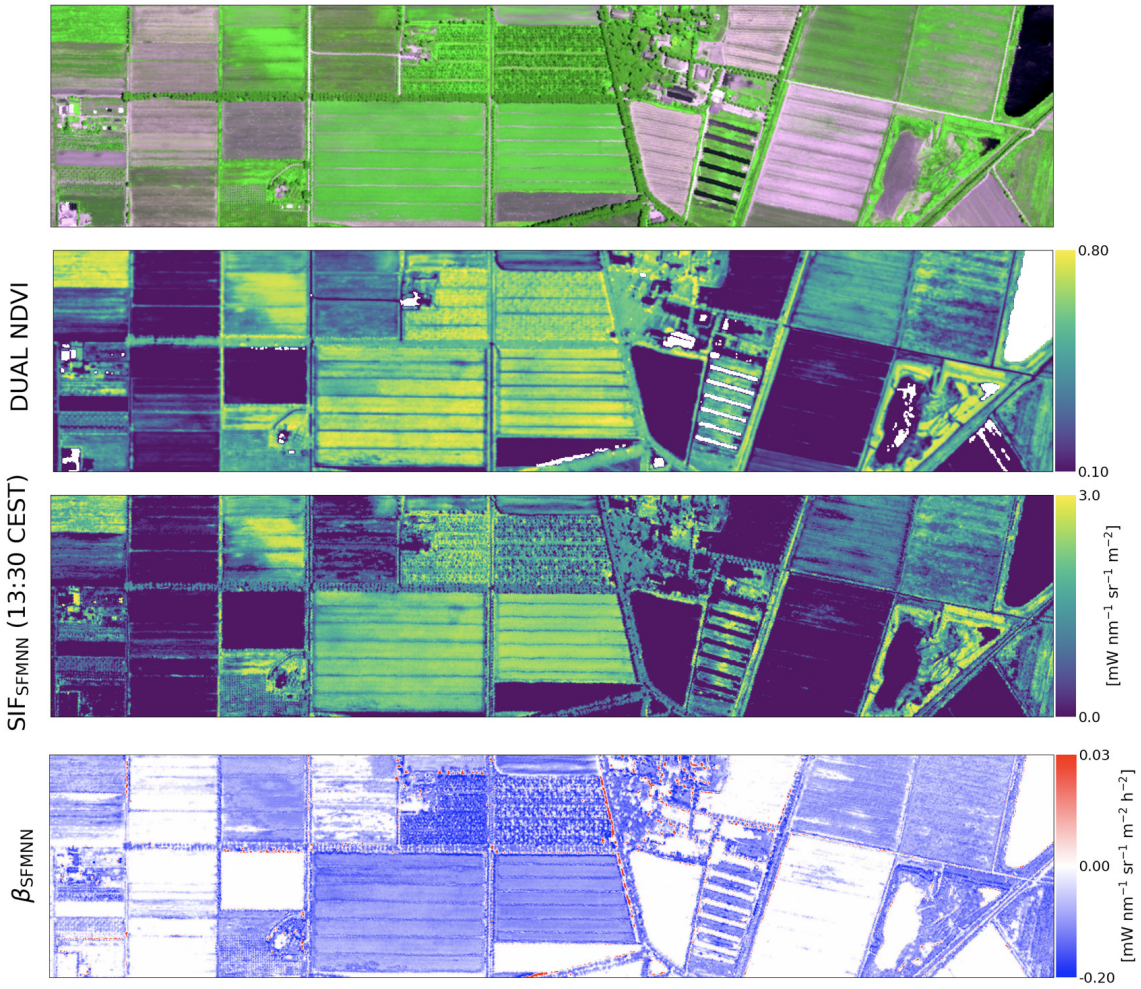


Figure 8: We show a pseudo-color image of a WST-2019 (1500 m) acquisition (first row), NDVI (second row), SFMNN SIF prediction at 13:30 CEST (third row), diurnal SFMNN SIF variation parameterized by β (fourth row). Pixels with invalid NDVI values (in shaded and barren pixels) are masked out.

fractional band depth

$$r_{O_2-A} = \frac{\min_\lambda t_{\text{tot}}(\lambda)}{\max_\lambda t_{\text{tot}}(\lambda)} \tag{21}$$

predicted by SFMNN for each spatial pixel. The optical path length d of any acquisition is computed from a DEM warped to the sensor geometry and the sensor flight height. Additionally, we take account of the varying downwelling viewing angle in each across-track pixel.

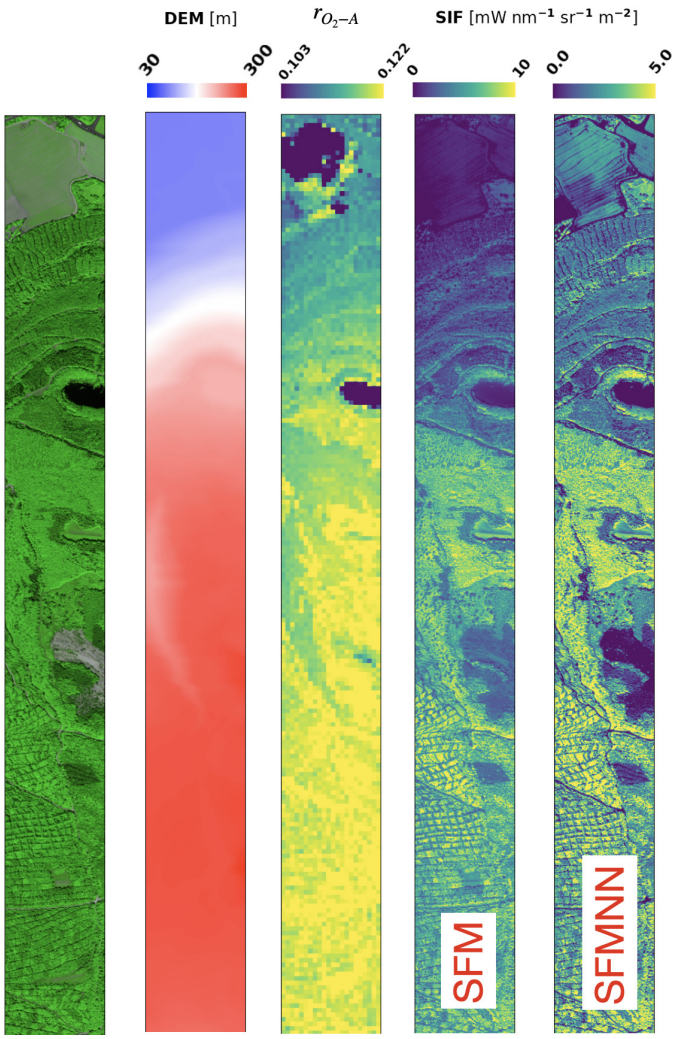


Figure 9: An exemplary airborne data acquisition with pronounced topographic variation (SOP-2023, 2023/06/23 14:44 CEST). We show a false color image of the acquisition, a digital elevation map in sensor geometry (DEM), the band depth r_{O_2-A}, the corresponding SFM SIF and SFMNN predictions.

In Fig. 9 we show an exemplary acquisition whose predicted band depth r_{O_2-A}varies along a topographic gradient with a height difference of $\sim$ 270 m. A mean difference of > 5 mW m^{-2} sr^{-1} nm^{-1} can be observed along this gradient in the SFM-SIF product, in which topographical changes have not been taken into account. The SFMNN predictor adapts the predicted band depth locally. As a result we can observe a constant SFMNN SIF prediction along-track for the different height ranges.

The predicted O_2-A band depth is not completely decoupled from surface related quantities. This can be seen in the exemplary acquisition plot in Fig. 9 in some correlated features in SIF in r_{O_2-A}. We can not determine from observational data whether this coupling is physical (i.e. conditioned by common causal factors) or conditioned by a learned empirical distribution.

In order to assess whether the variation in predicted band depth r_{O_2-A} is consistent across multiple acquisitions and across different predictors, in Fig. 10 we show the linear extinction factor $s_{r_{O_2-A}}$ defined as the first-order derivative to the approximation

$$\mathbb{E}\left[r_{O_2-A} \,|\, d\right] = s_{r_{O_2-A}} \cdot d + \text{const.} \tag{22}$$

where d denotes the geometrical optical path between the surface and sensor. As we are interested in the sensitivity of r_{O_2-A} to the changing optical path, we summarize any other influencing factor under *const*.

We find that $s_{r_{O_2-A}}$ stabilizes at a constant value in acquisitions with large differences in optical path lengths $\Delta d = \max d - \min d$. This is true for acquisitions of different SFMNN models and across different data sets. As expected, the variation in the $s_{r_{O_2-A}}$ estimates increases when the optical path length differences Δd become smaller and the uncertainty on the per-acquisition expectation $\mathbb{E}\left[r_{O_2-A} \,|\, d\right]$ increases.

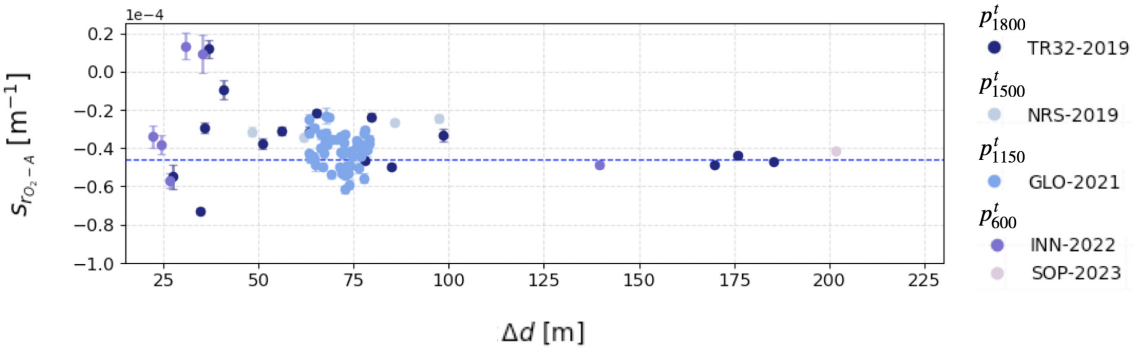


Figure 10: Estimated $s_{r_{O_2-A}}$ for all acquisitions with relevant variation in topography (cf. *topo* in Tab. 2). The dashed line represents the mean of four acquisitions with largest Δd.

5. Discussion

5.1. Reconstruction Performance

Since the SFM is the operational SIF retrieval algorithm for HyPlant data, its prediction as well as reconstruction performance has been repeatedly validated using top-of-canopy spectral measurements [36, 43, 53] and cross-validated with SIF products from different platforms [71]. Matching SFM's reconstruction performance is thus an important feature of novel spectral fitting algorithms aiming to retrieve SIF. While a good reconstruction performance does not validate the physicality of our approach, it constitutes a necessary precondition for any spectral fitting method. Failure to satisfactorily approximate the model to observations could indicate either an incomplete model formulation or too stringent constraints imposed as regularization on the fitting process.

We could show that the reconstruction performance of SFMNN is comparable to SFM in a data set of nine HyPlant acquisitions recorded over the course of a single day. SFMNN's reconstruction led to smaller RMSE than SFM both inside and outside of the O_2-A absorption band. We hypothesize that the SNR-based loss weighting, which increases the importance of residuals within the absorption band, led to this preferential improvement of absolute residuals as no such prior assumption is considered in the spectral fitting process of SFM. Parallelly, the improved reconstruction performance of SFMNN could be observed when analyzing its radiance normalized residuals nRMSE. Although SFMNNs nRMSE is smaller, we found the same differentiation in nRMSE between *noon* and *off-noon* conditions for SFMNN as for SFM; significantly worse nRMSE are found in *off-noon* conditions. This performance deterioration could be established in SFMNN residual statistics of multiple acquisitions (cf. Fig. 5). While we could not yet pinpoint the exact reasons for this behaviour, we posit that there are two plausible reasons.

First, the worse performance at low SZA may be due to data quality deterioration in lower light conditions. At lower SZA the diffusive contribution to the at-sensor radiance increases strongly even under clear-sky conditions when HyPlant data is normally recorded. An increasing diffusive component, however, reduces the signal-to-noise ratio of the at-sensor radiance as the pixelwise hyperspectral signal is spatially mixed. This reduces the ability of the network to disentangle fluorescence from reflectance and atmospheric components the at-sensor signal is composed of.

Secondly, there might be increasing model incompatibility under high SZA conditions. The physical model used for signal reconstruction in Eq. 1 includes a a diffusive contribution to the at-sensor signal. However, it is not clear if the PCA-based atmospheric transfer parameterization proposed in this work indeed yields physically

valid spectra under various observational conditions. Moreover, the proposed loss formulation might introduce varying prediction biases under changing observational conditions. The lack of control over the atmospheric decomposition is a significant draw-back of the PCA-based parameterization as compared to a physically based prediction of atmospheric components. This work has only addressed the atmospheric at-sensor contribution in an aggregated way by analyzing the variation of the O_2-A band depth under topographic changes. Further work should therefore evaluate quantitatively the predicted atmospheric components and concentrate on proposing physical constraints regarding the disentangling of individual atmospheric components.

A promising possibility to constrain further the simulated atmospheric transfer consists in replacing the PCA-based forward simulation in SFMNN with a differentiable emulator of a physically accurate radiative transfer model. Research in biophysical modelling of sun-induced fluorescence [72] and simulation of hyperspectral at-sensor radiance on various platforms [70, 73, 72, 74] are the basis for fast and accurate emulation models [75, 76]. The differentiable nature of a number of those emulator models warrants an investigation into possibilities to implement such a replacement that would implement a tight integration of physically and statistically based SIF retrieval. Furthermore, the use of sensor-specific emulation would facilitate the generalization of SFMNN to other airborne hyperspectral imaging sensors such as CFIS, as well as to spaceborne imaging sensors with appropriate spatial and spectral resolution, e.g. FLEX and DESIS. With the use of emulators the radiative transfer physics and sensor characteristics affecting the at-sensor radiance can be modularized by domain experts and do not need to be considered by the network architecture.

5.2. Accuracy of the SFMNN top-of-canopy SIF product

Direct validation with high-quality predictions from top-of-canopy measurements is the gold-standard for any SIF retrieval scheme if the spatial resolution of the sensor data is high enough to resolve the measurement footprint of the top-of-canopy measurements. The acquisition of field data and its synchronization with sensor acquisitions is, however, costly and usually results in small and spatially very constrained validation data sets. In this work we have gathered the majority of FLOX derived top-of-canopy SIF estimates acquired in tandem with HyPlant flights covering the period 2018-2022 to thoroughly validate SFMNN with high-quality data. While this study is the largest and most complete systematic comparison of HyPlant derived SIF estimates with top-of-canopy measurements, the validation data set has

still a limited size and mainly covers crop canopies.

In our validation study we could show that SFMNN prediction resulted in MAEs covering the range from 0.2 - 0.7 mW nm^{-1} sr^{-1} m^{-2} depending on the campaign data set used for validation. The FLEX mission requirement for SIF retrieval accuracy of 0.2 mW nm^{-1} sr^{-1} m^{-2} [73] could not be consistently achieved for HyPlant data with the current set-up using one of the three SIF retrieval methods (iFLD, SFM, SFMNN) that were investigated in this study.

5.3. Learning of implicit constraints

We have shown in two application studies that SFMNN predictions are constrained implicitly. The observation of a constraint on the second order derivative $\beta > 0$ in diurnal SIF products suggests that the network is able to learn a first order physiological relationship between day time (i.e. SZA) and SIF emission magnitude. On the other hand, it could be shown that the atmospheric transfer function could be adapted in a way to represent the variation of the optical path in acquisitions solely based on radiance data. The fact that the feature based internal representations learned by SFMNN can be shown to be sensitive to basic physiological and physical constraints emphasizes that the proposed loss formulation allows the network to generalize the disentangling beyond a pure functional optimization as it is done in SFM. This aspect of SFMNN can be considered to be an important advantage of SFMNN over the methodology followed by SFM as it allows in principle the use of trained models on data that was not included in the training. If a single model could be applied directly to new data prediction times could be significantly reduced as compared to the approach followed here (pretraining and fine-tuning) as well as compared to related spectral fitting methods such as SFM. Further validation should therefore especially concentrate on the generalization capacity of SFMNN SIF prediction for an operational context where fast prediction times are advantageous.

The feature representation of the hyperspectral signal as learned by the SFMNN encoder is based on a restricted number of HyPlant acquisitions. Recent scientific breakthroughs in the domain of neural network training for vision applications draw however on the self-supervised, application-agnostic learning of feature representations from very large data sets. Recently, first adaptations of large Vision Transformer and Diffusion models have been developed for remote sensing RGB [77, 78] and multi-spectral imagery [79]. The multitude of spaceborne, globally distributed hyperspectral datasets that have become and will be available in the near future (PRISMA [80], EnMAP [81], DESIS, FLEX, CHIME [82]) are likely to allow similar training set-ups. A natural extension of the present work will therefore consist in adapting the training of the SFMNN encoder backbone to a large collection of hyper-

spectral data sets from different sensors and platforms to improve its representative power.

6. Conclusion

We have presented in this work SFMNN, a self-supervised deep learning method to estimate SIF in the O_2-A absorption band of hyperspectral HyPlant imagery. We have applied the method to HyPlant acquisitions from multiple years and different observational conditions and compared the results to two SIF retrieval methods for HyPlant data by performing a validation with ground based FLOX measurements. This is the first time that HyPlant SIF retrieval methods and products have been compared systematically in a validation with a FLOX in-situ data set spanning multiple years.

We could show that SFMNN yielded state-of-the-art SIF predictions similar to the SFM baseline. This could be observed both from the good performance when compared with in-situ data as well as the time series data matching well the expected diurnal SIF dynamics in vegetated and non-vegetated areas. We could furthermore observe that SFMNN could be applied to HyPlant data taken in regions with large topographic variation. One of the most serious issues of the baselines considered in this work is to find a suitable parameterization of the atmospheric transfer in regions with changing atmospheric and observational conditions. As SFMNN can locally compensate such changes in principle, an increased interest was put in its ability to yield trustworthy SIF estimates in such conditions. Since no in-situ data was available for flight lines exhibiting large topographic changes, we could only conduct a qualitative analysis. We found that the effective extinction coefficient determined from the linear relationship between O_2-A band depth and the estimated radiative path length converged to a constant value in flight lines with large topographic variations.

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Appendix A. Derivation of SNR-based loss weighting

Let us consider a simplified model for SIF estimation and assume we have N measurements y on two wavelengths λ_1 and λ_2. Signal is composed by a reflectance part $y_R(\lambda)$ and a fluorescence part $y_R(\lambda)$ including atmospheric influences, such that

$$y(\lambda) = y_R(\lambda) + y_f(\lambda) + \epsilon \tag{A.1}$$

We model $R(\lambda)$ and $f(\lambda)$ to be

$$R(\lambda_i) = c_R(\lambda_i) R_0 \tag{A.2}$$
$$f(\lambda_i) = c_f(\lambda_i) f_0 \tag{A.3}$$

with known, constant function $c_\cdot(\lambda)$ and scalar parameters R_0 and f_0. In a more realistic model there would be more than one parameter for R and f. Please note, that we can select $c_\cdot(\lambda)$ such that $\mathbb{E}[R_0] = 1$ and $\mathbb{E}[f_0] = 1$ if we want to. Our model equations for estimation are then

$$y(\lambda_1)_j = A_R(\lambda_1) c_R(\lambda_1) R_0 + A_f(\lambda_1) c_f(\lambda_1) f_0 + \epsilon_j \tag{A.4}$$
$$y(\lambda_2)_j = A_R(\lambda_2) c_R(\lambda_2) R_0 + A_f(\lambda_2) c_f(\lambda_2) f_0 + \epsilon_j \tag{A.5}$$

where A_R and A_f contain all atmospheric parameters and all the rest. Note, that the multiple measurements indicated by index j only change ϵ_j. Obviously, we can multiply each equation with a freely adaptable weight w_i without changing anything. We can rewrite the equation to be

$$\mathbf{W}\mathbf{y} = \mathbf{W}\mathbf{X}\beta + \mathbf{W}\epsilon \tag{A.6}$$

where $\mathbf{W}$ is a diagonal weight matrix with $W_{ii} = w_i$, $X_{i1} = A_R(\lambda_i) c_R(\lambda_i)$, $X_{i2} = A_f(\lambda_i) c_f(\lambda_i)$, $\beta_1 = R_0$, and $\beta_2 = f_0$. Noise ϵ has $\mathbb{E}(\epsilon|\mathbf{X}) = 0$ and $\mathrm{cov}(\epsilon|\mathbf{X}) = \mathbf{\Omega}$. In our case we assume that noise is independent, such that $\mathbf{\Omega}$ is diagonal with $\Omega_{ii} = w_i^2 \sigma_i^2$, where σ_i is the noise standard deviation for wavelength i. Clearly, weights w_i also scale the noise in the weighted equations. We know from Gauss-Markov-Theorem that the generalized least-squares estimator can then be written as

$$\hat{\beta} = (\mathbf{X}^T \mathbf{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{\Omega}^{-1} \mathbf{y} \tag{A.7}$$

where weights w_i cancel out. Using this simultaneous estimation of all parameters β_i for designing a suitable loss function is not obvious. We therefore investigate a special case, where only one parameter is unknown.

Let us assume R_0 to be known and we want to infer f_0. We can then reformulate (A.4) as

$$\tilde{\mathbf{y}} = \mathbf{X} f_0 + \epsilon \tag{A.8}$$

where $\tilde{y}_i = y_i - y_{R,i}$ and $X_i = A_f(\lambda_i) c_f(\lambda_i)$. The BLUE is given by

$$\hat{f}_0 = \frac{\sum_i X_i y_i / \sigma_i^2}{\sum_i X_i^2 / \sigma_i^2} \tag{A.9}$$

or, more suggestively written

$$\hat{f}_0 = \sum_i \frac{1}{\sigma_i^2} \frac{X_i}{\sum_j X_j^2 / \sigma_j^2} \tilde{y}_i \tag{A.10}$$

For the usual Moore-Penrose pseudo-inverse we would get

$$\hat{f}_0 = \sum_i \frac{X_i}{\sum_j X_j^2} \tilde{y}_i \tag{A.11}$$

The difference between these two estimation approaches is in the weights

$$w_i = \frac{1}{\sigma_i^2} \frac{\sum_j X_j^2}{\sum_j X_j^2 / \sigma_j^2} \tag{A.12}$$

For training, we derive a mean weighting function $w_f = \mathbb{E}_p[w_i(L_{\mathrm{at-s}}, f)]$ over a specific distribution p of simulated at-sensor radiances $L_{\mathrm{at-s}}$ and top-of-canopy fluorescence. The simulations are performed in this case with a simple two-stream model, a fixed atmospheric transfer function and varying reflectance and fluorescence functions. Given the HyPlant-specific sensor variance model σ^2, this expectation can be written as

$$w_f(\lambda) = \left\langle \frac{1}{\sigma_\lambda^2} \frac{\sum_{\lambda'} f_{\lambda'}^2}{\sum_{\lambda'} f_{\lambda'}^2 / \sigma_{\lambda'}^2} \right\rangle_{p(L_{\mathrm{at-s}}, f)} . \tag{A.13}$$

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